monolingual text. However, given the typically short snippets of conversation that may not have sufficient context, machine translation approaches may not always be effective. Work has been done on training language models for code switched text. An interesting question would be to consider how we can combine multiple language models such that we can handle code switched text effectively while not losing performance on normal, monolingual text as well. In addition to code switched text, we also need to handle transliterated text and mis-spelled text. This becomes all the more important when the targeted users can span different educational levels.

Now let us get to discussing how we can design the task-oriented dialogue system itself. Since the conversational assistant needs to support different tasks/domains such as restaurant booking, ticket reservation, calendar, reminders, etc, we need to build a domain classification system first to identify the domain/task targeted by the user utterance. From our design perspective, the conversational assistant needs to be extensible to handle a huge number of domains as it becomes popular, as is the case with virtual assistants such as Siri/Alexa/Google. How can we build a multi-domain classification system for utterances? Once we identify the domain, we need to have models for intent classification and entity identification for each domain. These models will be domain-specific.

One approach is to take the given utterance and run the domain-specific models for intent/entity for each domain. Whichever model scores the utterance with highest confidence, we assign the utterance to that domain and proceed further. However, this requires that we run the intent classifier and entity identification for all the domains for domain identification. Given that the system needs to support tens of thousands of domains, this approach becomes unscalable, since the intent classification and entity identification models are typically heavyweight. The alternative is to have a separate domain classifier first, which chooses 'K' most likely domains based on the utterance, and then for each of these K shortlisted domains, the intent and slot identification module can be run to obtain the classification results. Based on the intent classification and slot identification results along with certain contextual features, the target domain can be selected. This is known as the shortlister reranker based domain selection.

As we mentioned before, an intent classification problem can be treated as a short text classification problem. However, the main issue with each of these domain-specific intent classifier models is the dearth of training data. Typically, for each intent, there are only a handful of sample utterances, which makes the task of the intent classifier quite difficult, especially when you have intents that can be similar to each other. A good discussion of these issues is well covered in the recent paper 'HINT3: Raising the bar for Intent Detection in the Wild' by Gaurav Arora, *et al.* Interested readers should look up this paper for further details.

In our design, we also need to consider that users can come up with totally unrelated utterances in response to the virtual assistant, which can lead the conversation off its track. Also, since the virtual assistant can be used with users of different ages, there is a need to ensure hate speech or objectionable content is not supported as part of the dialogue. Hence, the dialogue system also needs to have an objectionable content classifier and flag a warning to the user in such a case, and finally terminate the conversation in case the warning is not heeded.

Another important design choice is to consider whether we should support only textual input or allow voice interactions as well. Voice as a medium has a much better adoption and ease of use. However, supporting voice input brings with it a whole new set of challenges. We need to have an automatic speech recognition (ASR) model, which again needs to have: (a) multilingual support, and (b) support code switched input. Errors by the ASR module can have a huge impact on the downstream NLP modules. Hence, post-processing error correction of the ASR output based on context is often critical. Support for voice interactions also requires that the assistant responses need to be converted back into natural sounding voice output.

Personalisation is an important factor in the design of a solution. Assume that our solution of a conversational assistant is deployed as a mobile app on a mobile device. If the same mobile device and app is being shared among a set of family members, personalisation becomes a daunting task. While there has been some work on supporting personalised interactions on a shared device, this area is not much explored yet.

As the solution gets more widely used, we need to be able to use feedback and data that can be used to retrain the various models and improve performance. An important question here is about collecting user feedback. Collecting explicit feedback from users gets us accurate information, but can be jarring to the user experience. Implicit feedback may not give an accurate signal for improving the system. It is important to talk about these trade-offs during the design interview. While you do not need to provide a solution to all the open issues, you should definitely highlight the important ones. I would recommend that readers who are preparing for system design interviews should practice (using a whiteboard) some of the common design interview questions available at: *https://github.com/donnemartin/system-design-primer#study-guide*.

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! Stay healthy and stay safe. END

By: Sandya Mannarswamy

The author is an expert in natural language processing and is currently working as an independent researcher. Her interests include natural language processing, machine learning and AI.